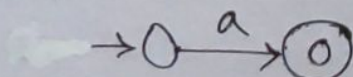
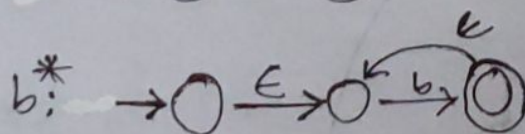
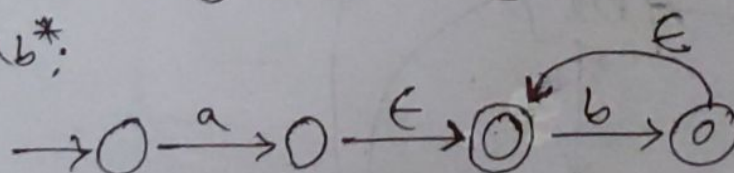
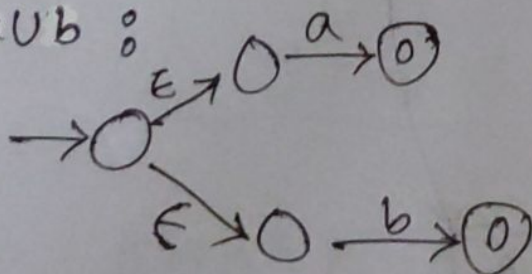
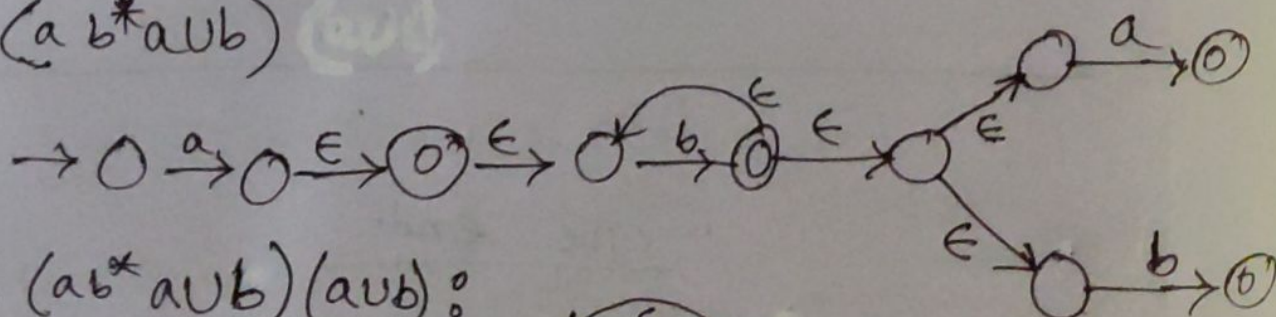
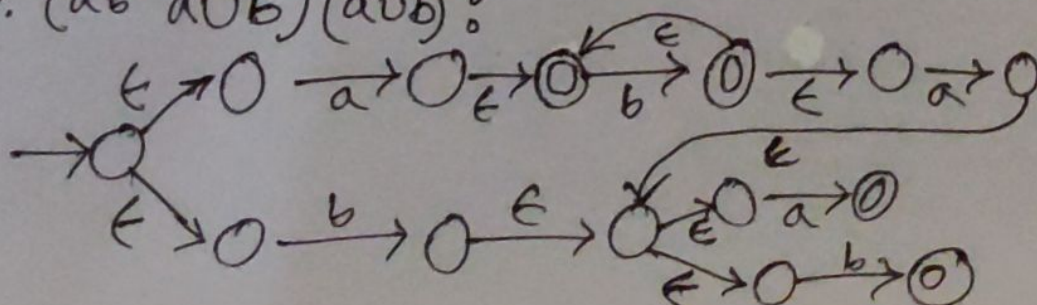


Answers to the ques no. 1 $(ab^*a \cup b)(a \cup b)$

a:

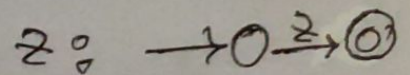
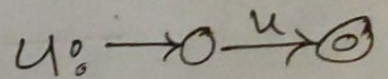
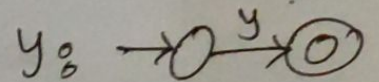
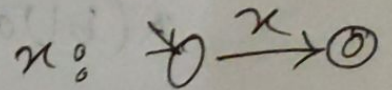
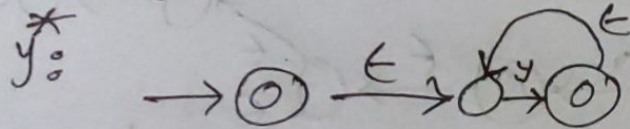


b:

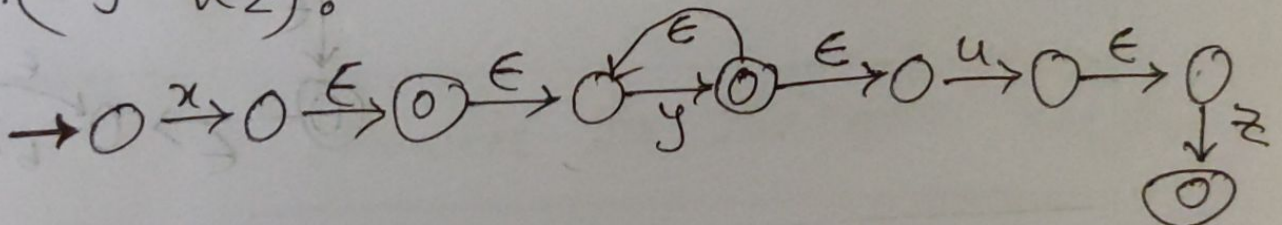
 b^* : ab^* : $a \cup b$: $(ab^*a \cup b)$  $\therefore (ab^*a \cup b)(a \cup b)$:

Answer to the ques no: 02

xy^*uz



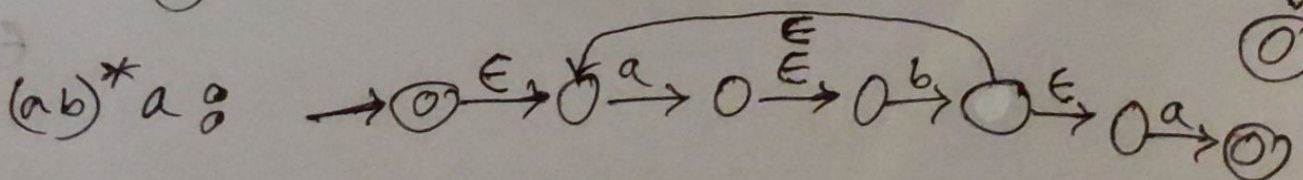
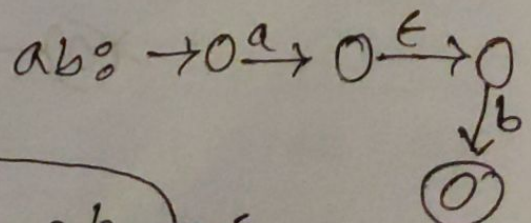
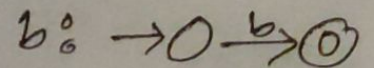
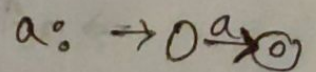
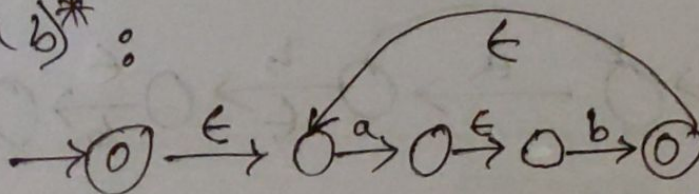
$\therefore (xy^*uz)$:



Answer to the ques: 03

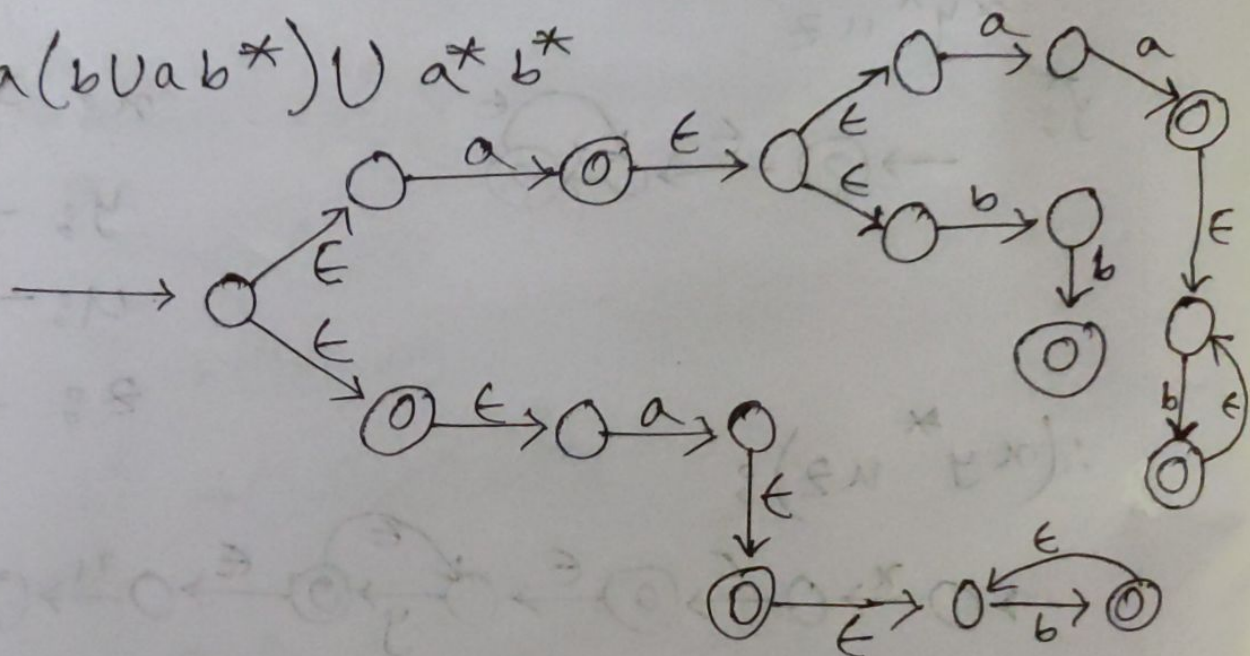
$(ab)^*a$

$(ab)^*$:



Answer to the ques: 04

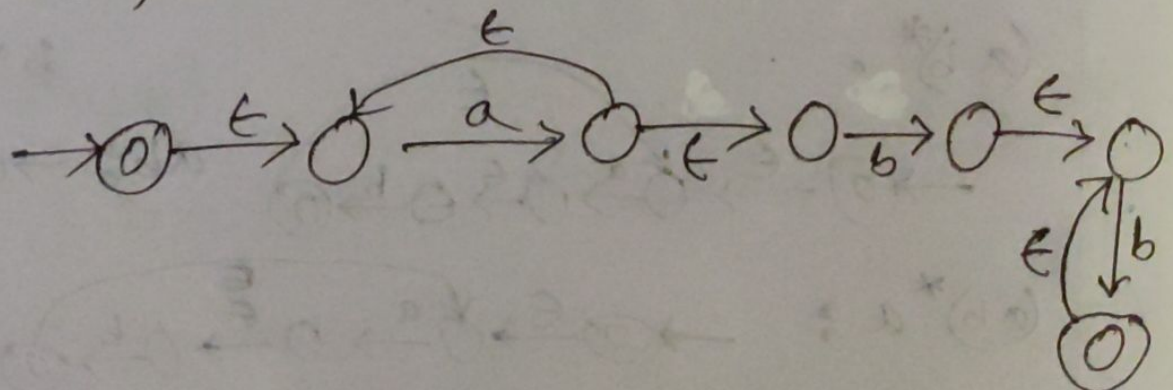
$$a(b \cup a b^*) \cup a^* b^*$$



Answer to the ques: 05

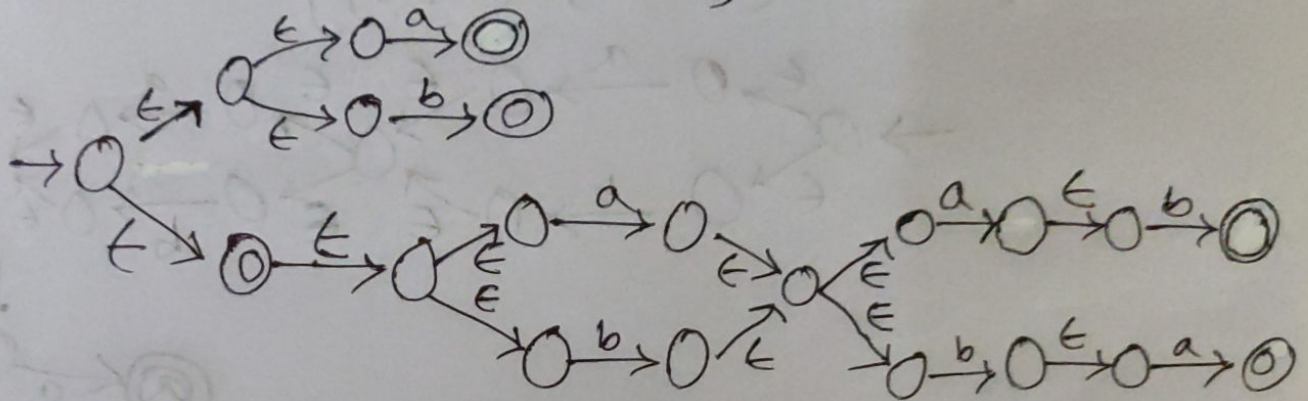
$$a^* b^+; b^+ [1 \text{ or more } b]$$

$$\Rightarrow a^* b b^*; b^+ [0 \text{ or more } b]$$



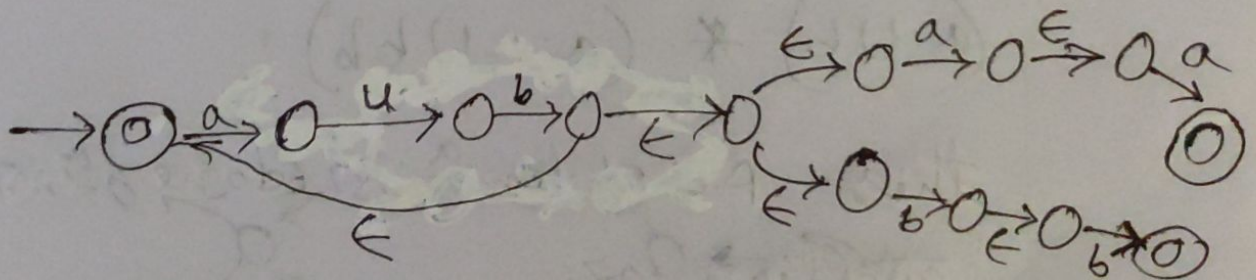
Answer to the ques: 06

$$a \cup b \cup (a \cup b)^* (a \cup b a)$$



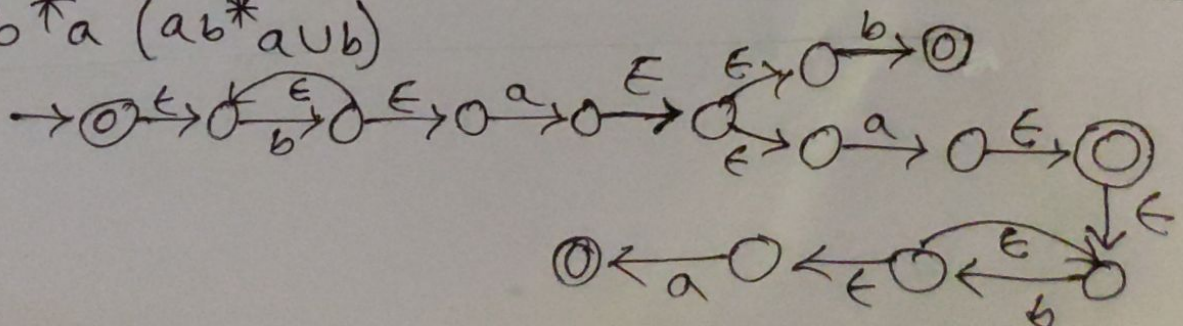
Answer to the ques no: 07

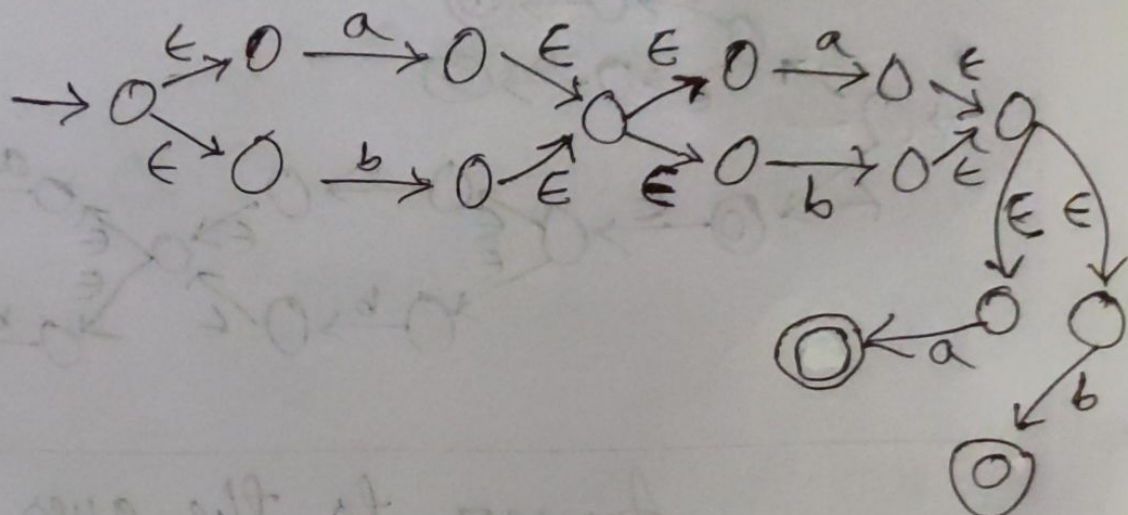
$$(a \cup b)^* (a a \cup b b)$$



Answer to the ques: 08

$$b^* a (a b^* a \cup b)$$



Answer to the ques no: 09. $(a \cup b) \ ((a \cup b) \ (a \cup b))$ Answers to the ques: 10 $(a \cup b) * (a \cup a \cup b \cup b) :$ 